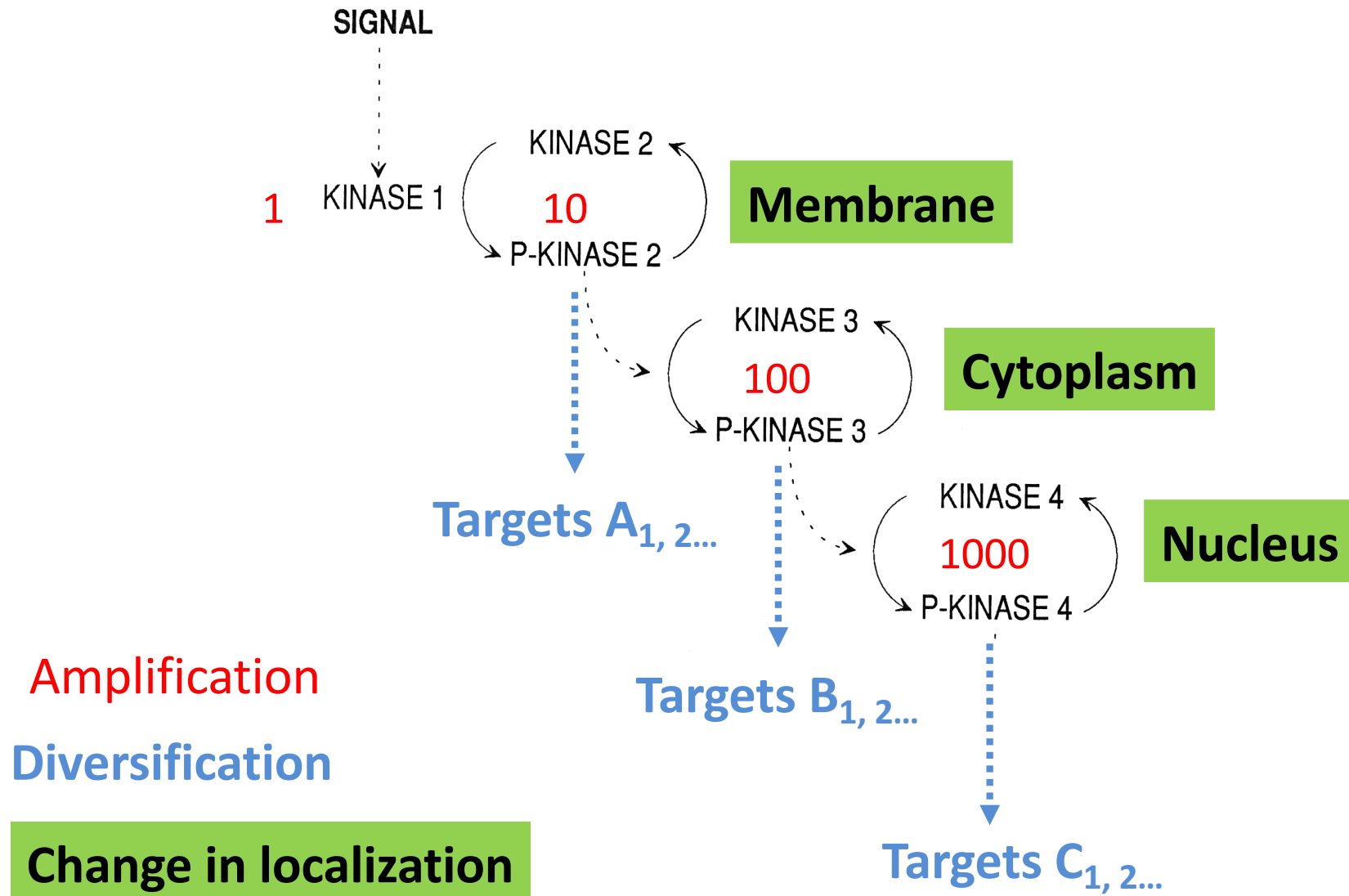


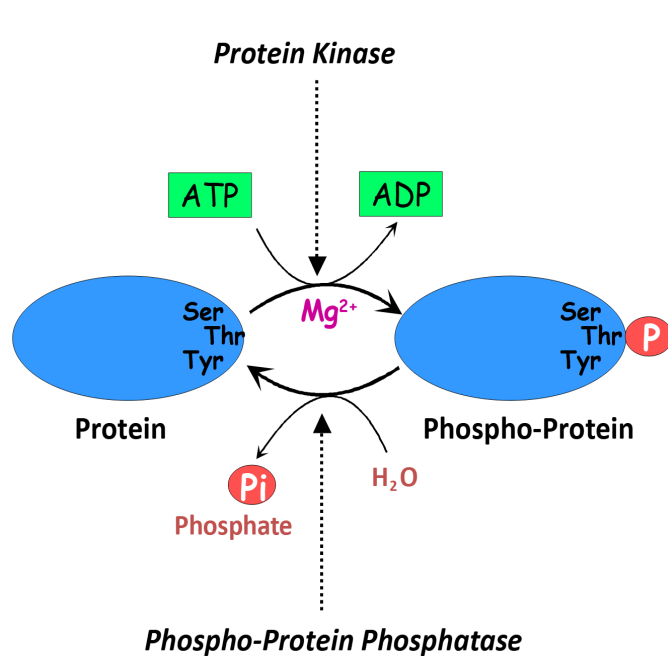
# MAP KINASE SIGNALING

MAP = Mitogen-activated protein

## SCHEME AND CONCEPTS FOR PHOSPHORYLATION CASCADES

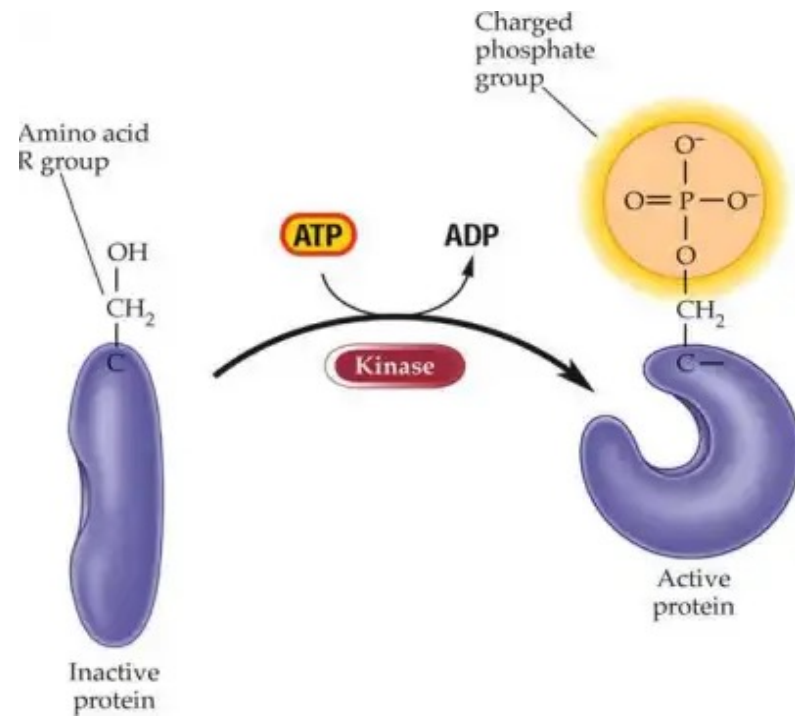


## PHOSPHORYLATION ON SERINE, THREONINE OR TYROSINE



Phosphoryl acceptor : Ser, Thr, Tyr

Phosphoryl donor : ATP

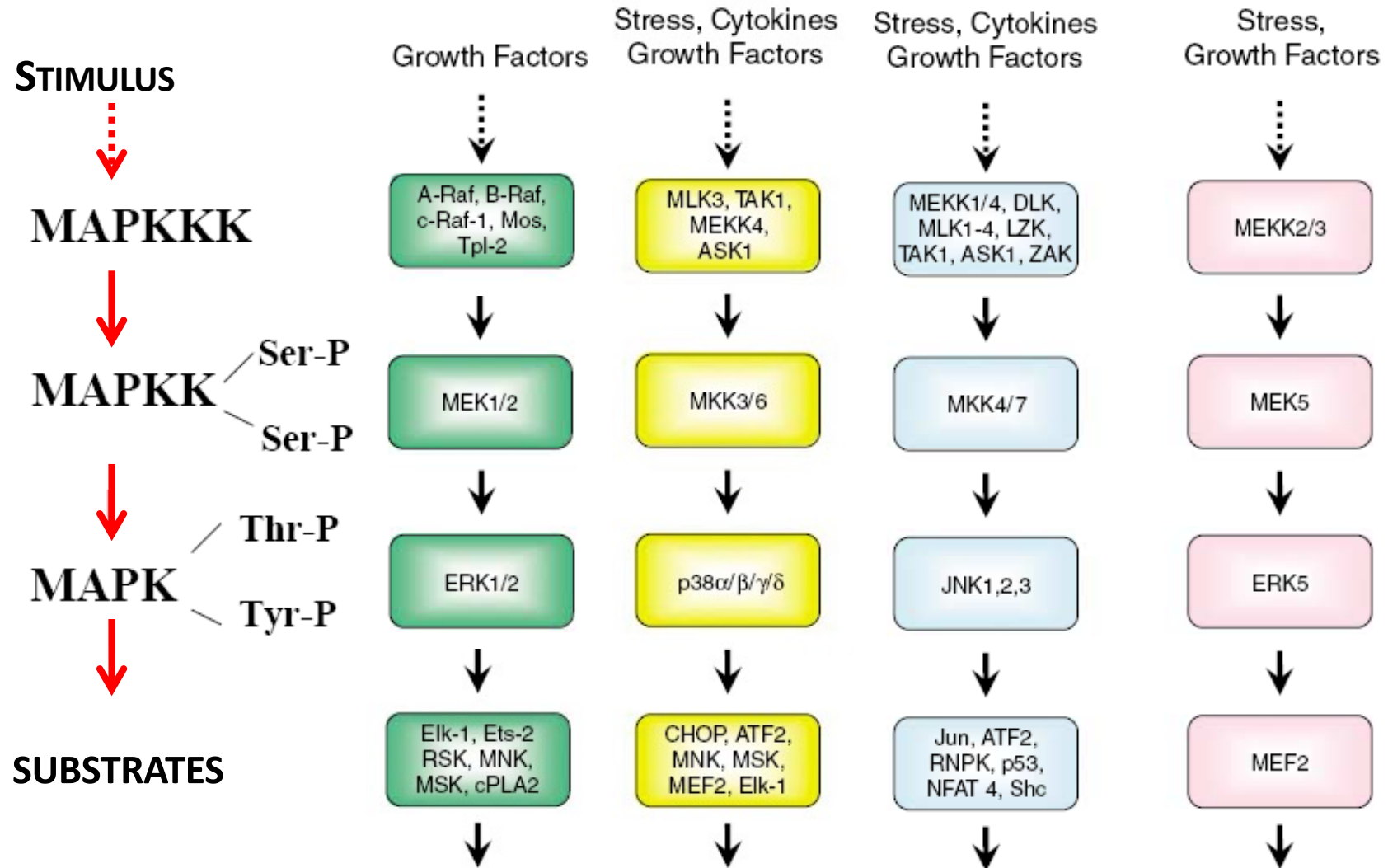


Phosphorylation is a **transient** event

**Phosphorylation induces a conformational change** that can modify :

Protein protein interaction  
Stability  
Activity  
Subcellular localization

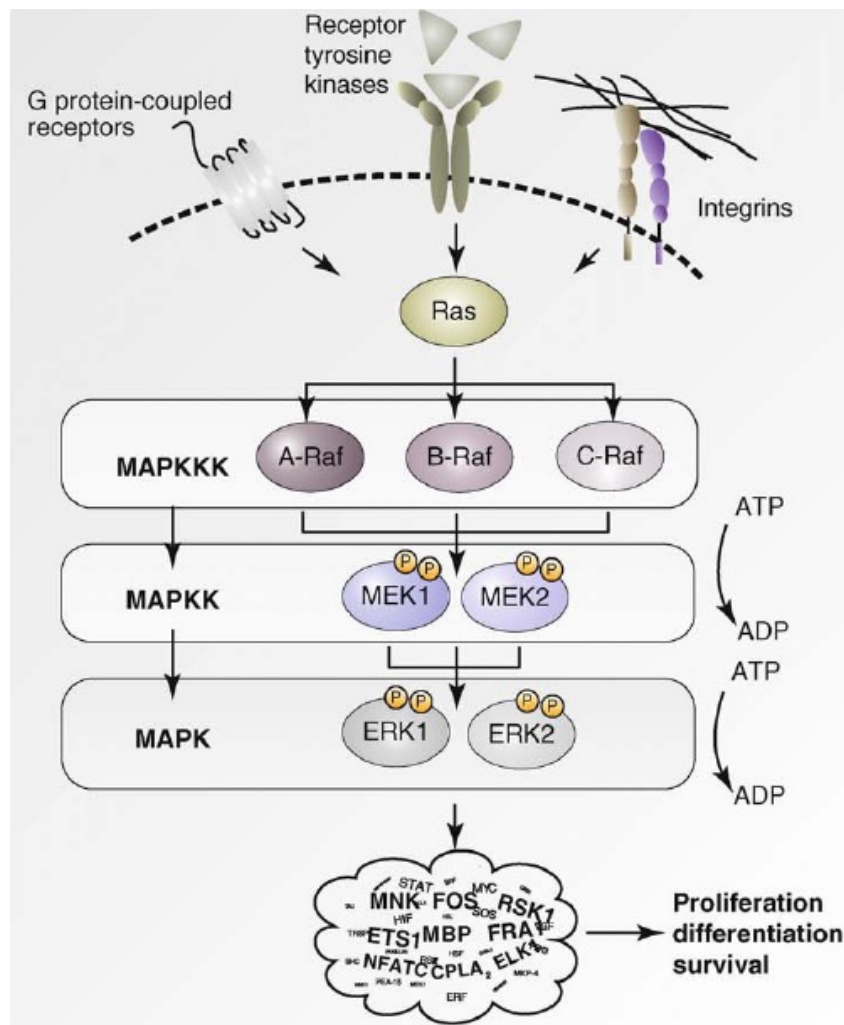
## MAPK: MITOGEN ACTIVATED PROTEIN KINASE



MAPKinase pathway can be initiated downstream several receptors and downstream extracellular matrix protein receptors integrins

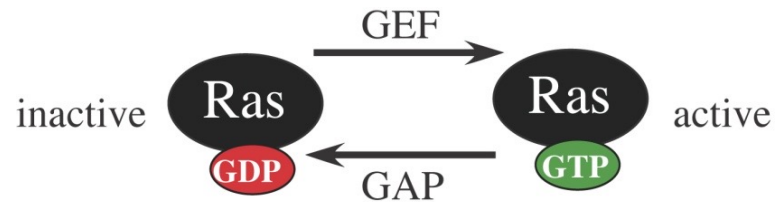
The first main step is activation of **Ras, a monomeric G protein**

Ras is a family of protein, which include RasGRF, a Ras activated by calcium.  
Thus MAPKinase can be activated by calcium as well (not shown on the scheme)



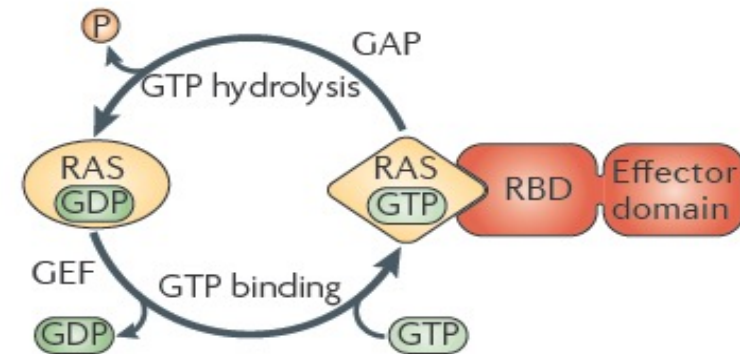
## Principle of monomeric G protein activation

GEF/ Guanine nucleotide Exchange Factor



GAP: GTPase Activating Protein

« activated » **monomeric G proteins** means the binding **with GTP** allows the protein to **interact with a protein on the RBD -Ras Binding Domain-** to induce **conformational change** that **modifies the activity** of that target protein.



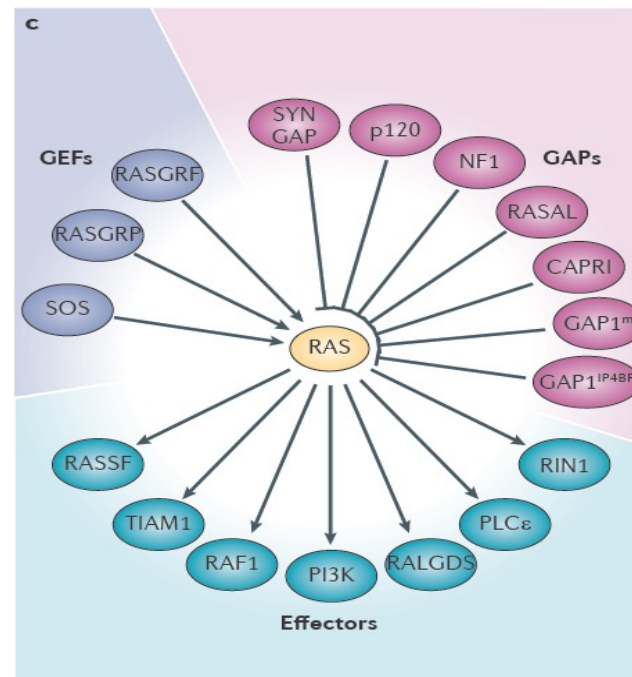
**Raf, the MAPKKK, has a RBD**

## Multiplicity in cell signaling

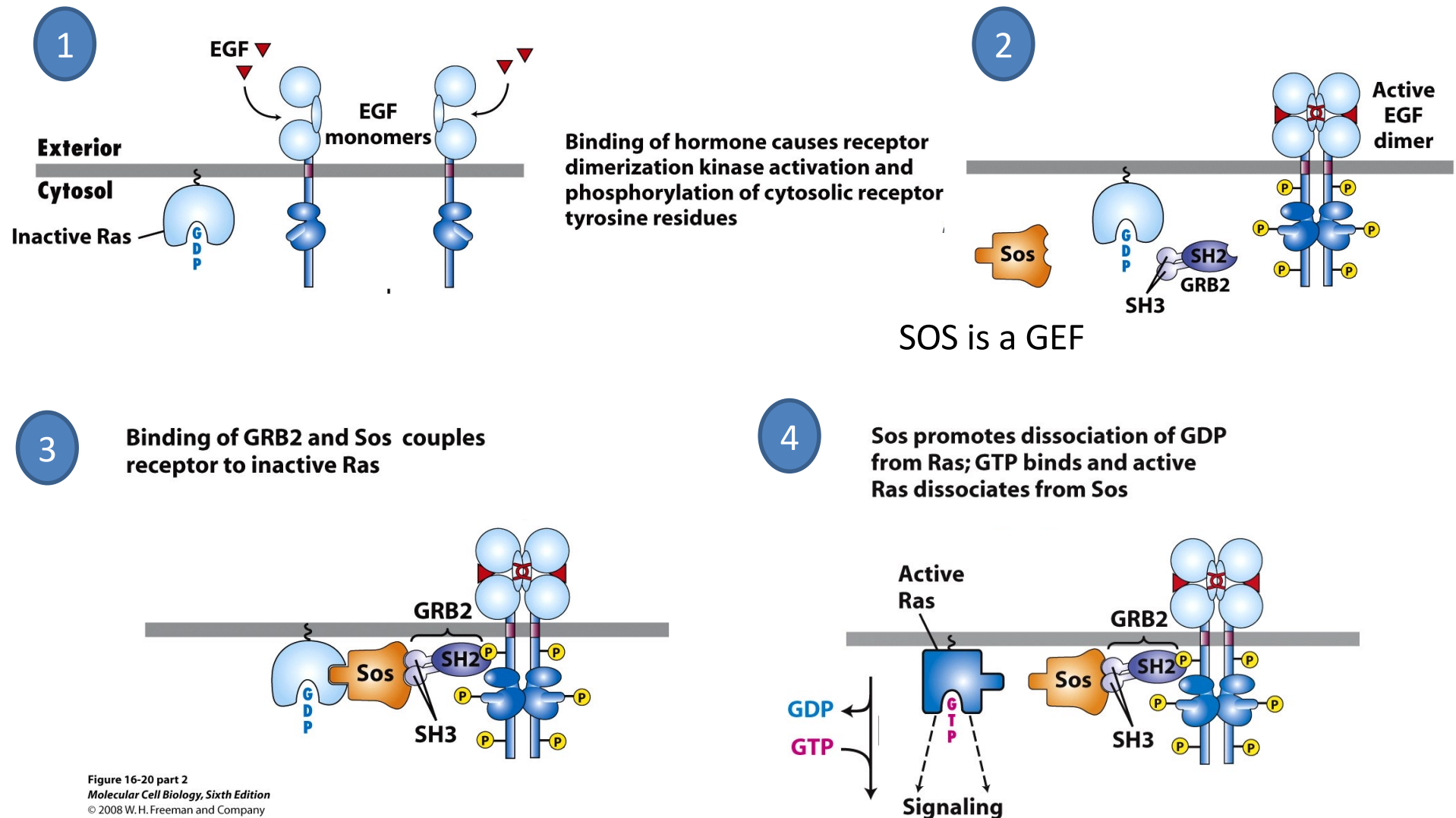
*Fine tuning of cell functions*

There are different GEF and GAP for Ras

And Ras activates several pathways



## Activation of MAPkinase pathway by RTK Receptor Tyrosine Kinase (growth factors receptors)

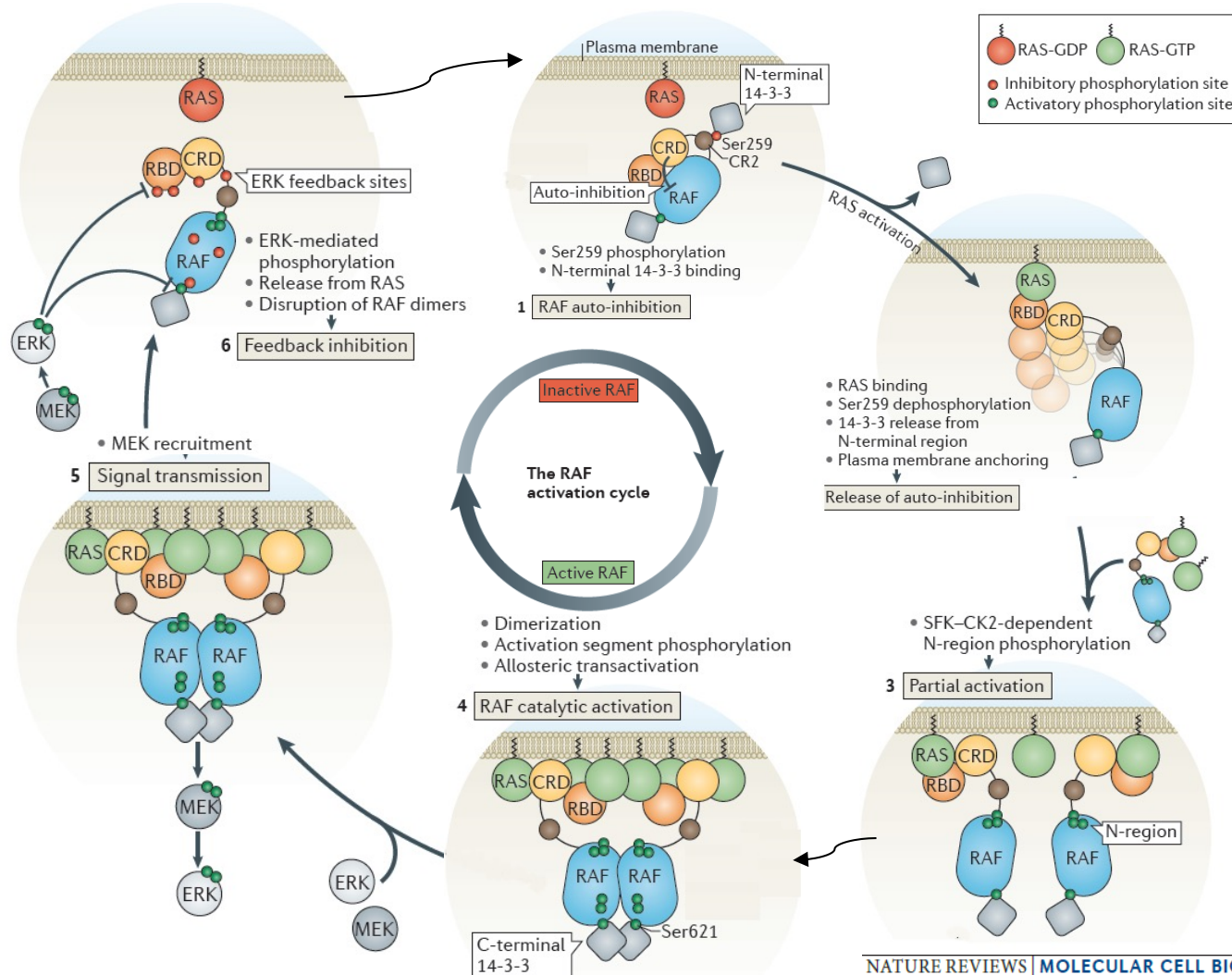


### Activation of RAF by Ras :

- **Interaction** with Ras induces **conformational change** of RAF to **modify its activity**
- Phosphorylation and dephosphorylation of RAF by itself and by other kinases and phosphatases

### Phosphorylations either activate or inactivate

- **Dimerization** of RAF for further conformational changes so RAF phosphorylates MEK
- **Negative feedback**, RAF activates MEK, which activates ERK, which inactivates RAF

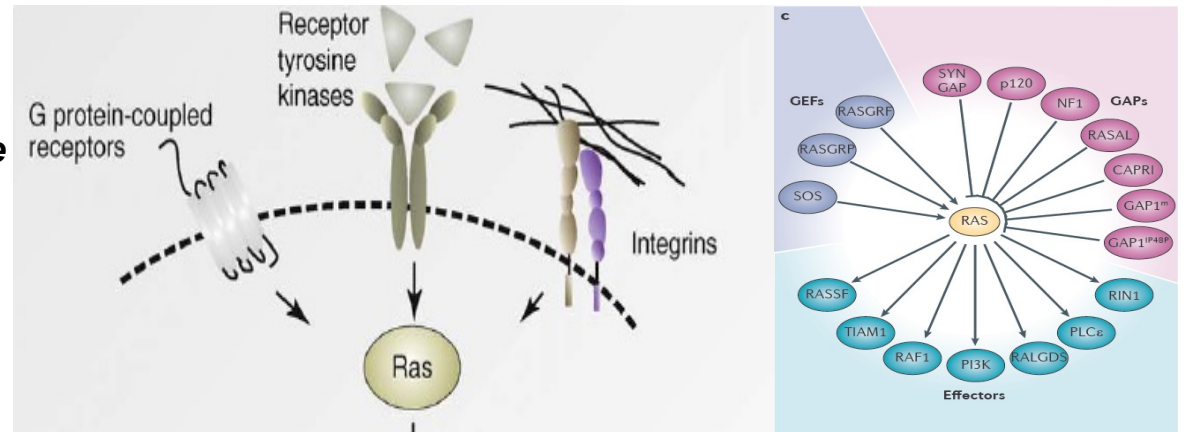


Multiplicity in cell signaling

***It is not happening everywhere all the time***

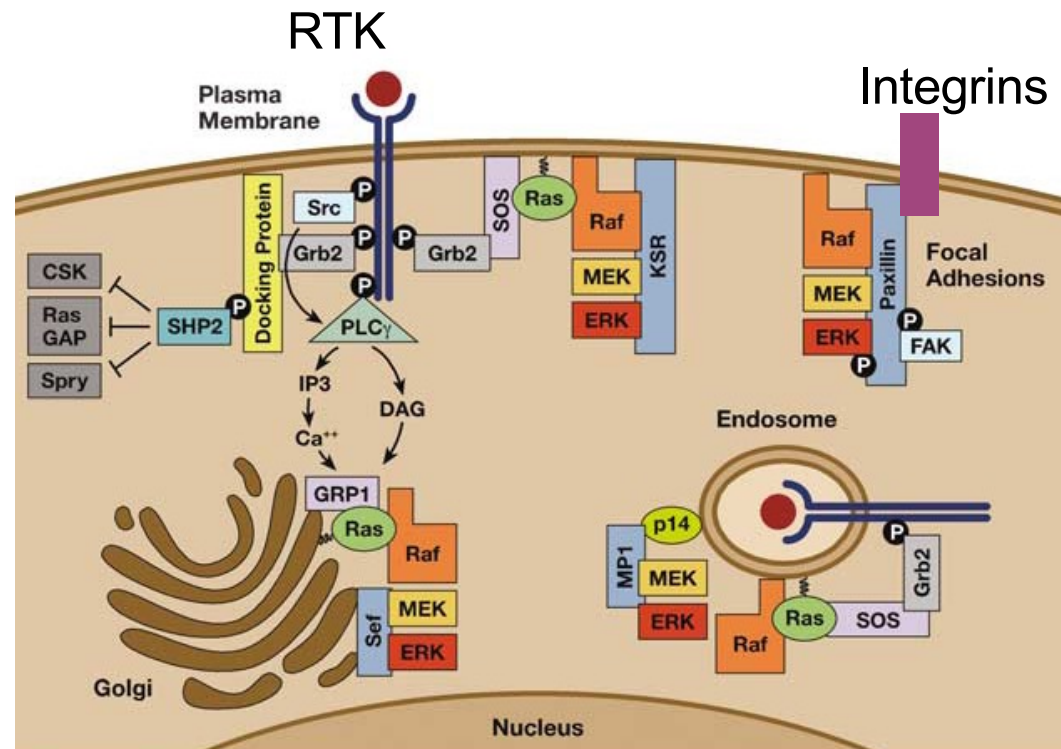
**Spatial localization is controlled**

Ras activated by RTK or by Integrins **do not induce same phenomena** in the cell



## SCAFFOLDING PROTEINS

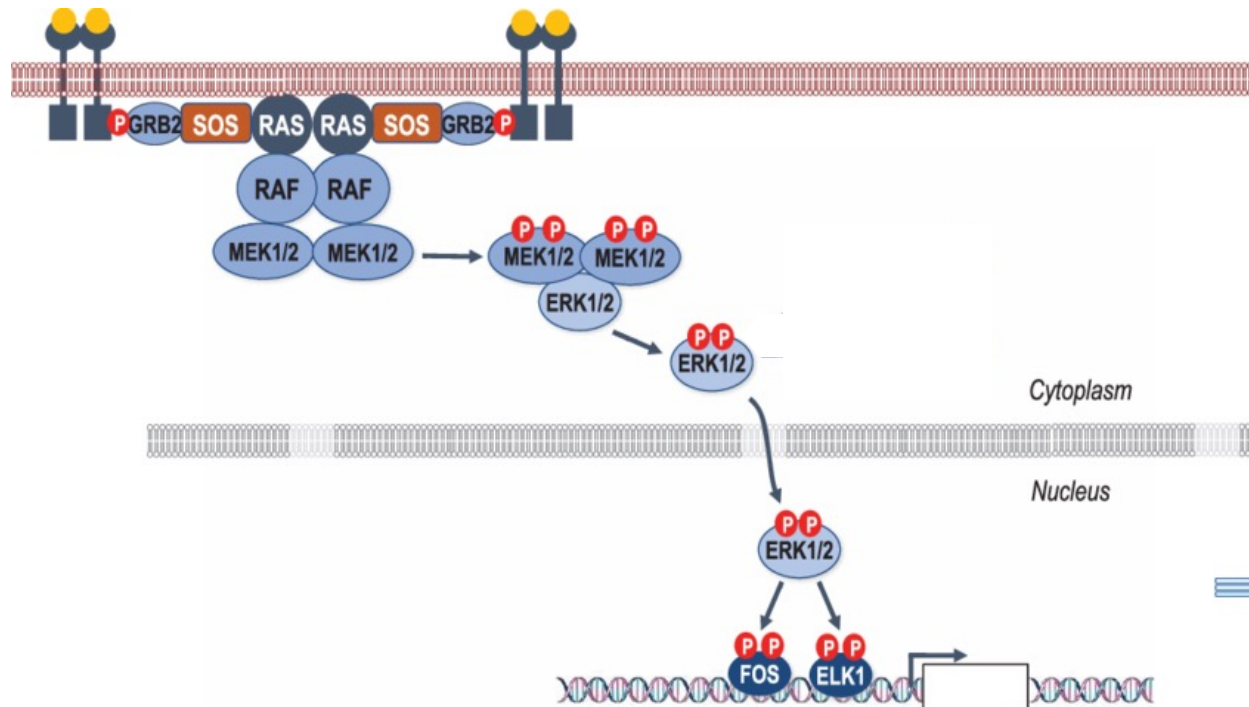
| ERK Scaffolds<br>(RTK Signaling) |                 |
|----------------------------------|-----------------|
| <b>KSR</b>                       | Plasma Membrane |
| <b>MP1</b>                       | Endosome        |
| <b>Sef</b>                       | Golgi           |
| <b>Paxillin</b>                  | Focal Adhesions |



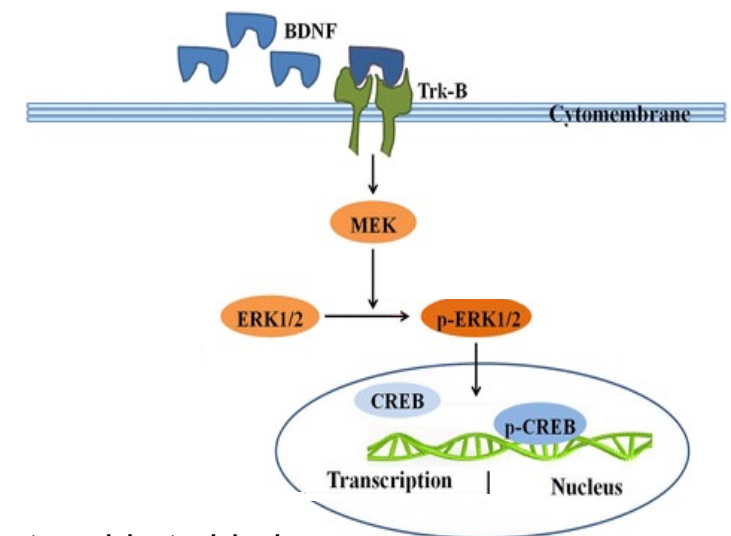
Phosphorylation of ERK instructs its **translocation** to the nucleus

The main function of ERK pathway is to **regulate gene expression**

**A brief extracellular signal induces a long-term cellular response**



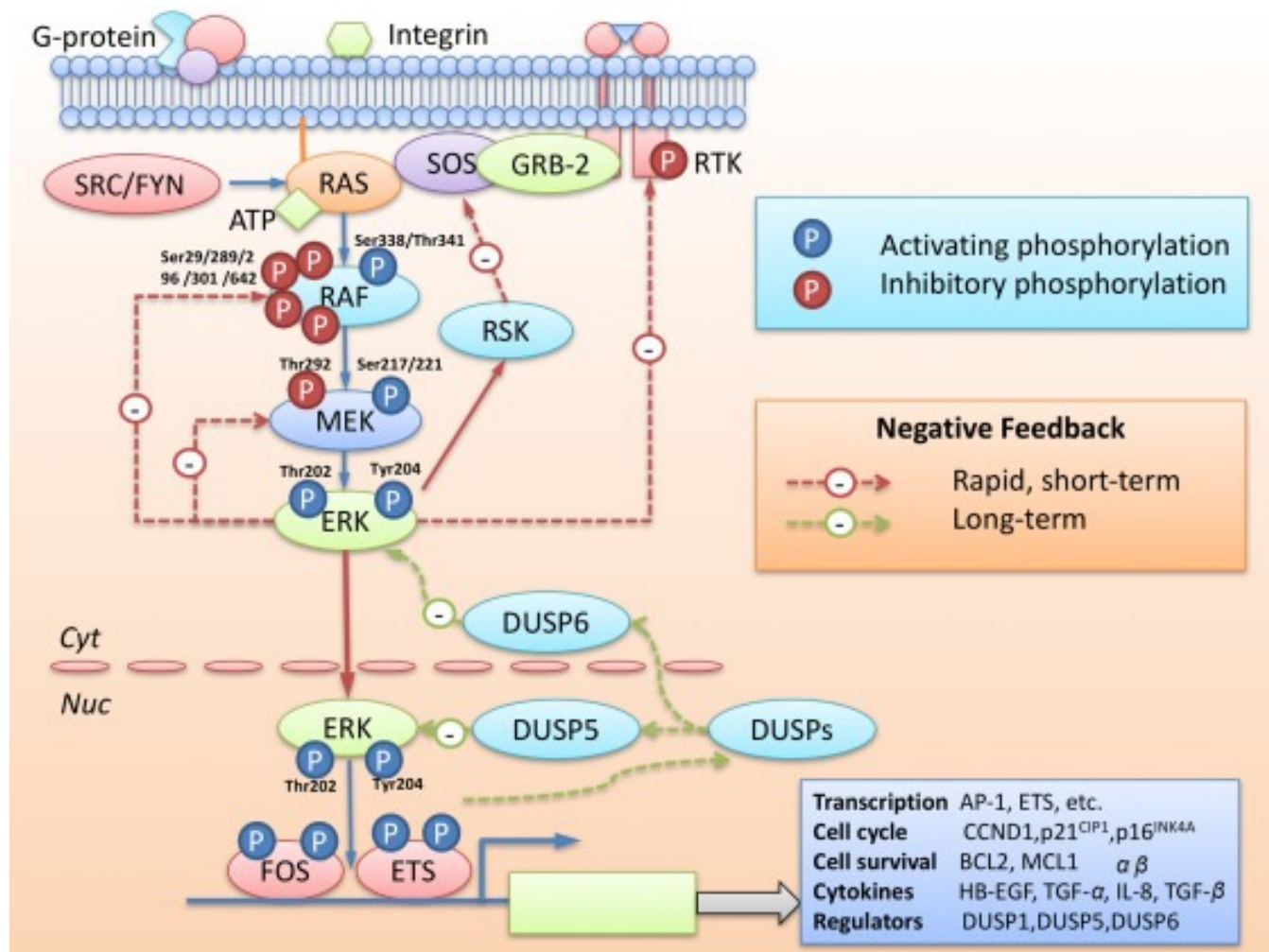
Fos and ELK1 are transcription factors  
They are activated by phosphorylation



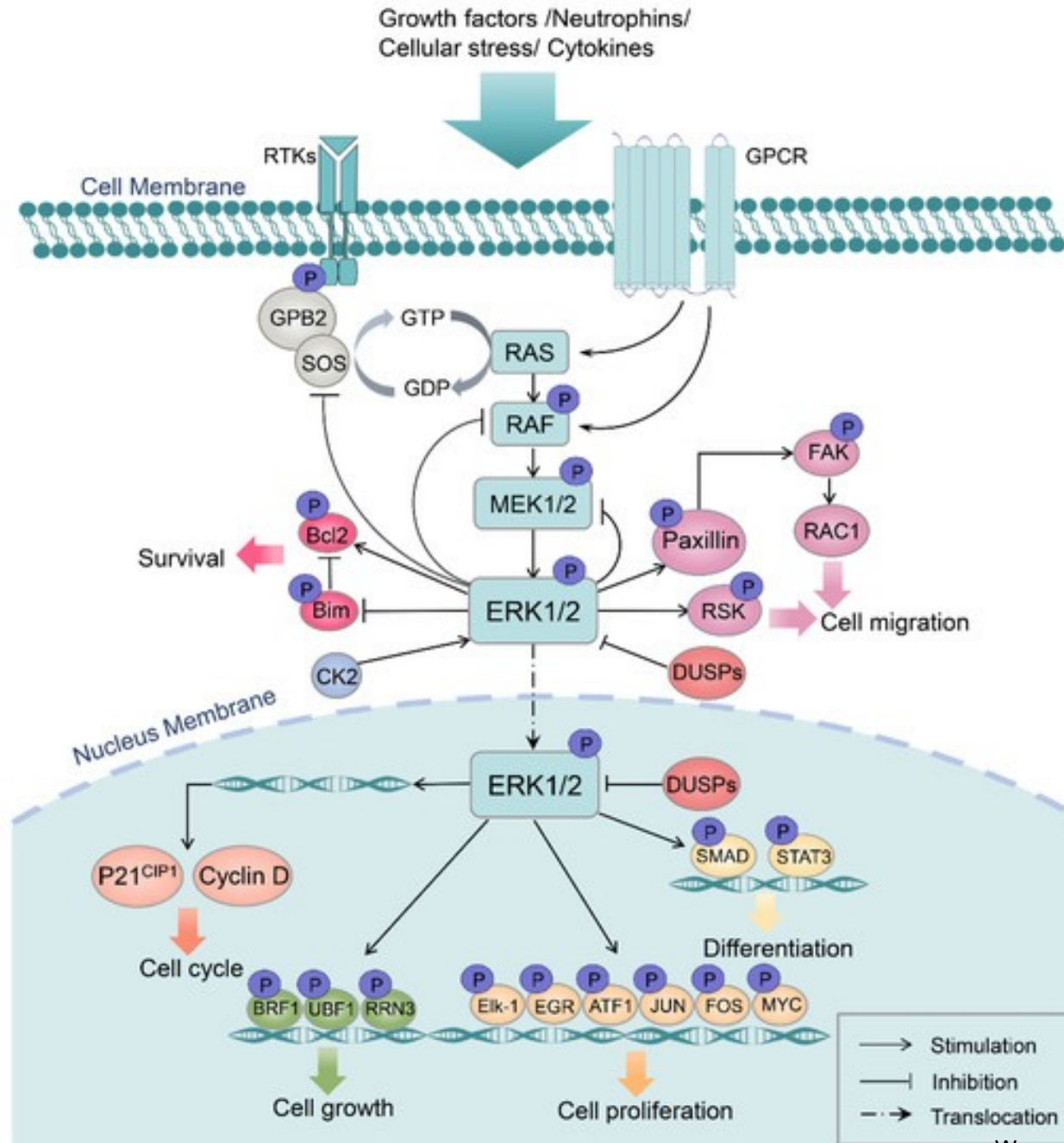
CREB (cAMP Response Element Binding protein): transcription factor able to bind, upon phosphorylation, CRE (DNA) sequences, hence promoting transcription of target genes

## Kinetics of cell signaling is controlled

Negative feedback by inhibitory phosphorylation  
Phosphatases



ERK pathway has several functions



ARTICLE **OPEN**

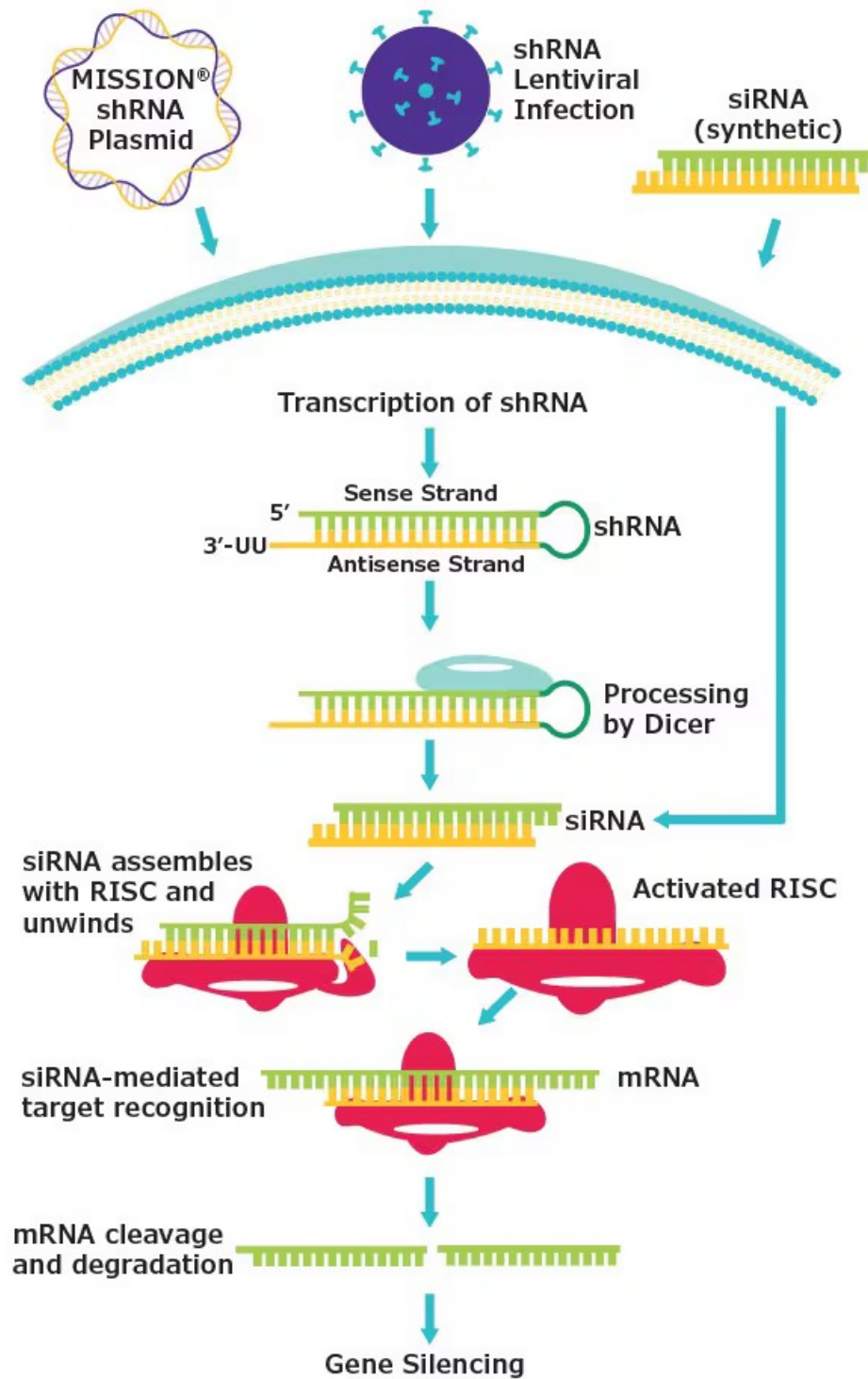


# Microglial ERK-NRBP1-CREB-BDNF signaling in sustained antidepressant actions of (*R*)-ketamine

Wei Yao<sup>1,7</sup>, Qianqian Cao<sup>2,7</sup>, Shilin Luo<sup>3,7</sup>, Lujuan He<sup>2</sup>, Chun Yang <sup>4,6</sup>, Jiaxu Chen <sup>1</sup>, Qi Qi <sup>5</sup>✉, Kenji Hashimoto <sup>4</sup>✉ and Ji-chun Zhang <sup>2</sup>✉

## Introduction to Methods:

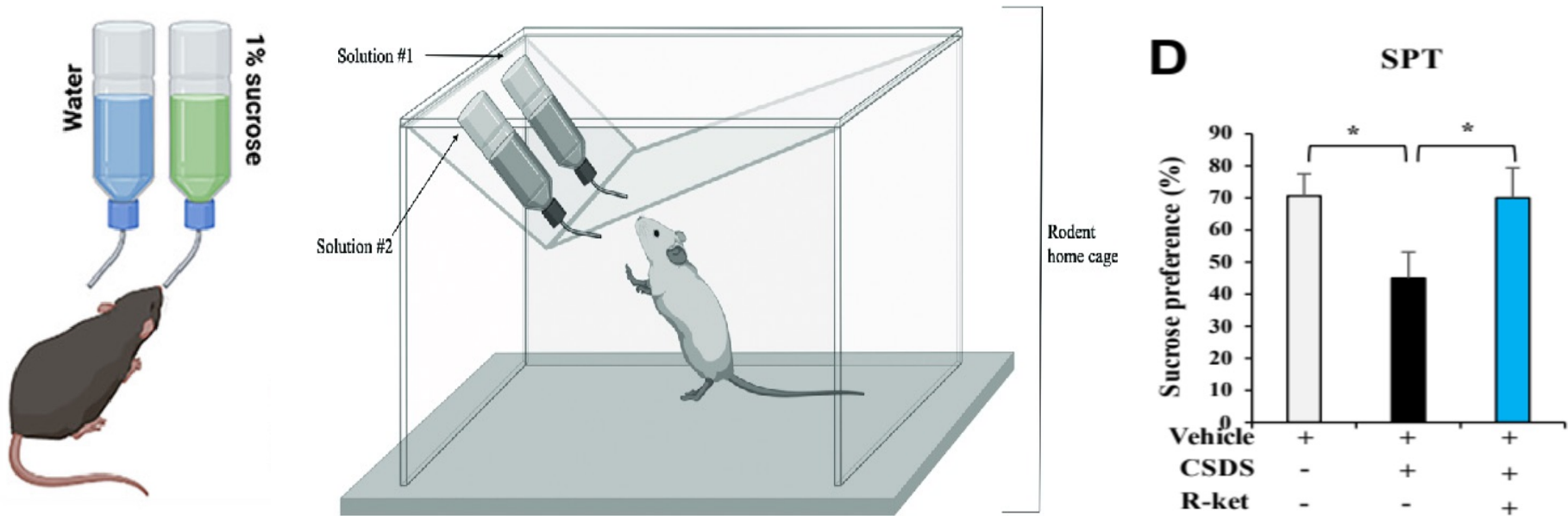
### 1. RNAi:



## 2. Sucrose Preference Test

**Anhedonia** (i.e. reduced motivation or ability to experience pleasure), a hallmark of depression, is evaluated with the sucrose preference test

Sucrose preference is measured with two bottle choice test

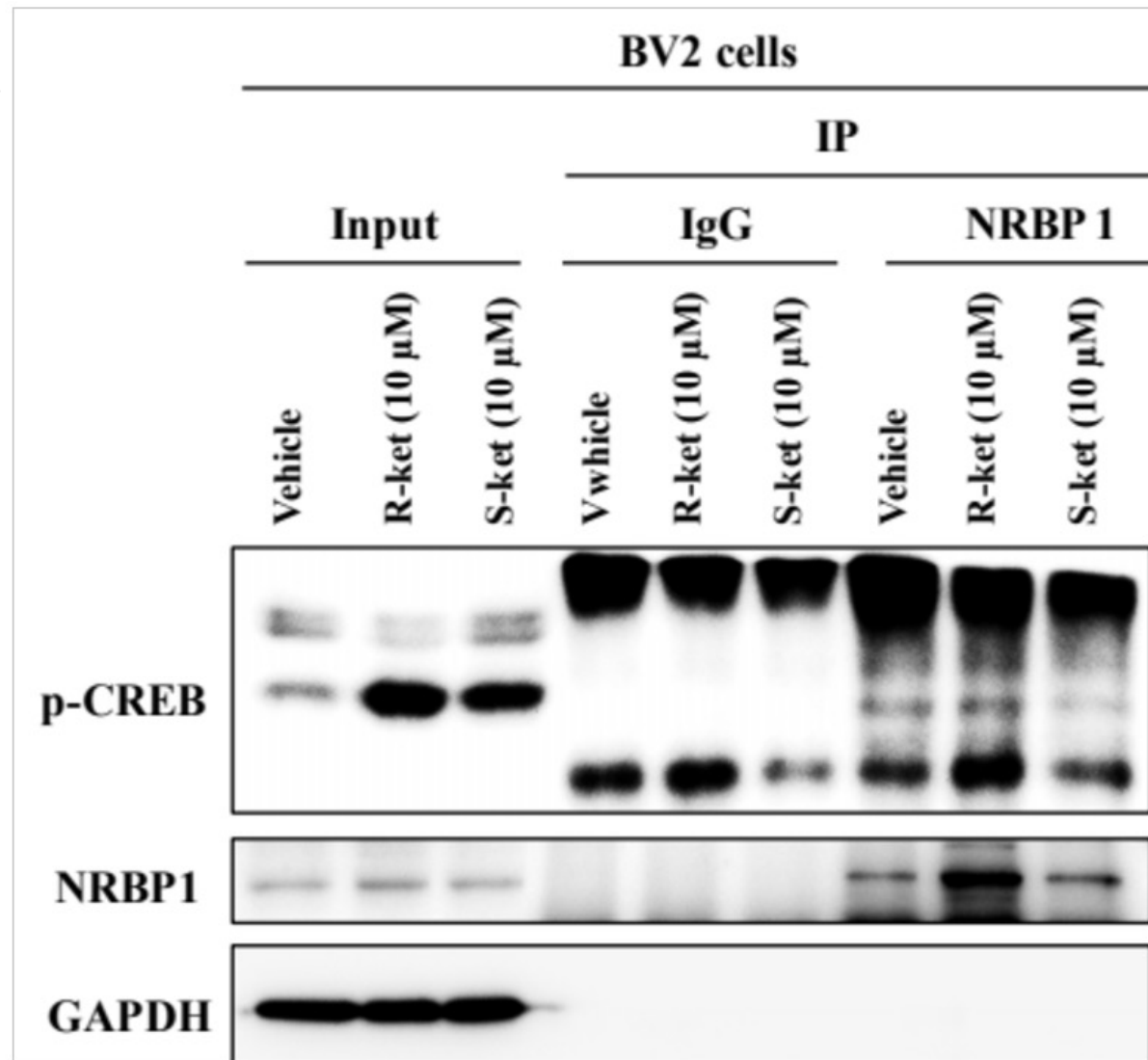


Mice were habituated to a 1% sucrose solution for 48 h before the test day. And then the mice were deprived of water and food for 4 h, followed by a preference test with water and 1% sucrose for 1 h. The bottles containing water and sucrose were weighed before and at the end of this period and the sucrose preference (%) was determined.

## Figure analysis:

1. Model
2. Conditions
3. Measurements (what, how..)
4. Observations
5. Interpretations
6. Question addressed
7. Perspectives

**Figure S2 GROUPE1**

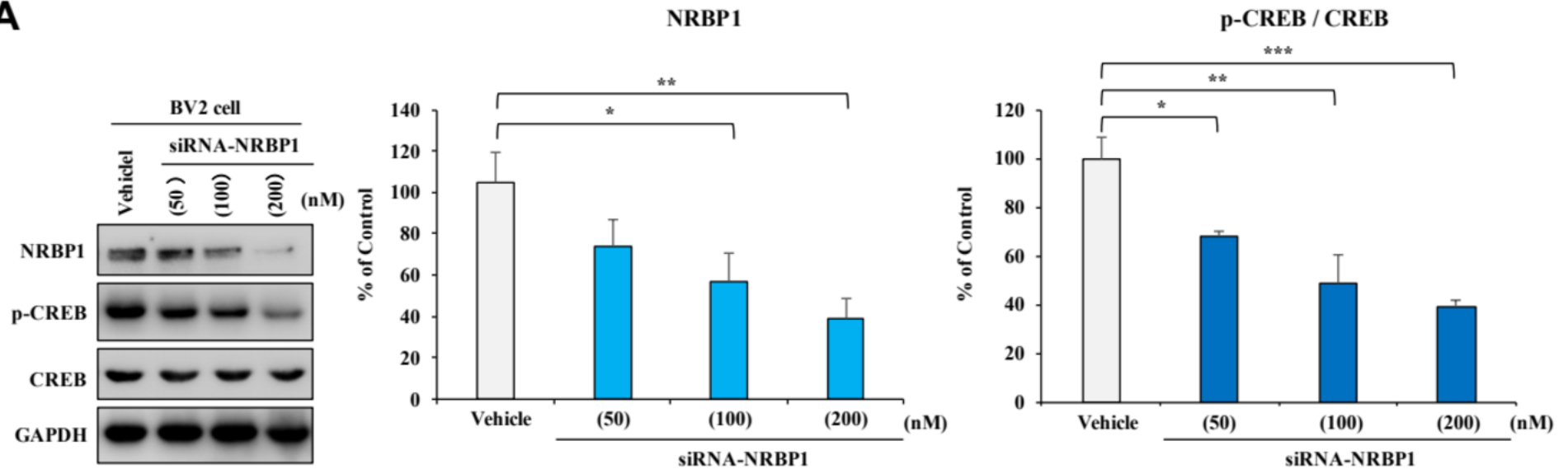


**Figure S2:** BV2 cells were treated with vehicle, (*R*)-ketamine (10  $\mu$ M) or (*S*)-ketamine (10  $\mu$ M) for 24 hours.

The BV2 cell line was obtained by immortalization of mouse brain microglial cells with a retrovirus carrying the v-raf/v-myc oncogenes. BV2 cells mimic the morphological (small cell body with branched processes) and functional characteristics of primary microglia, retaining their ability to produce cytokines and respond to stimuli like lipopolysaccharide (LPS) (inducing inflammation).

## Figure S3A GROUPE 2

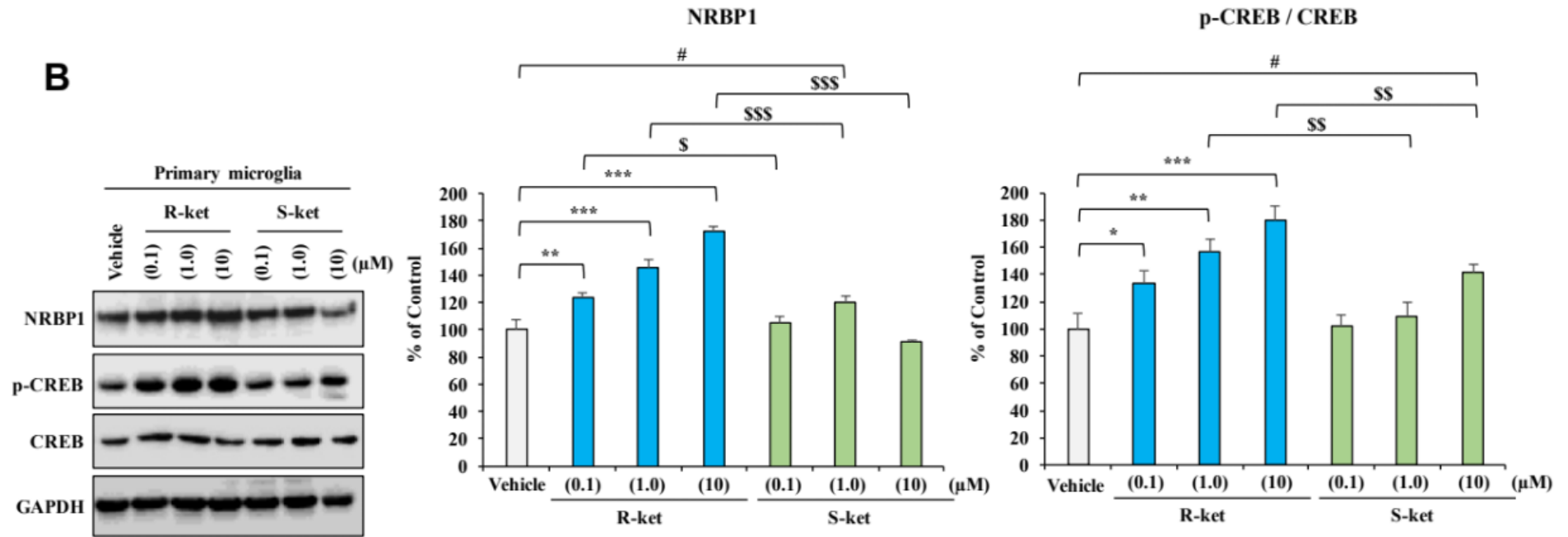
**A**



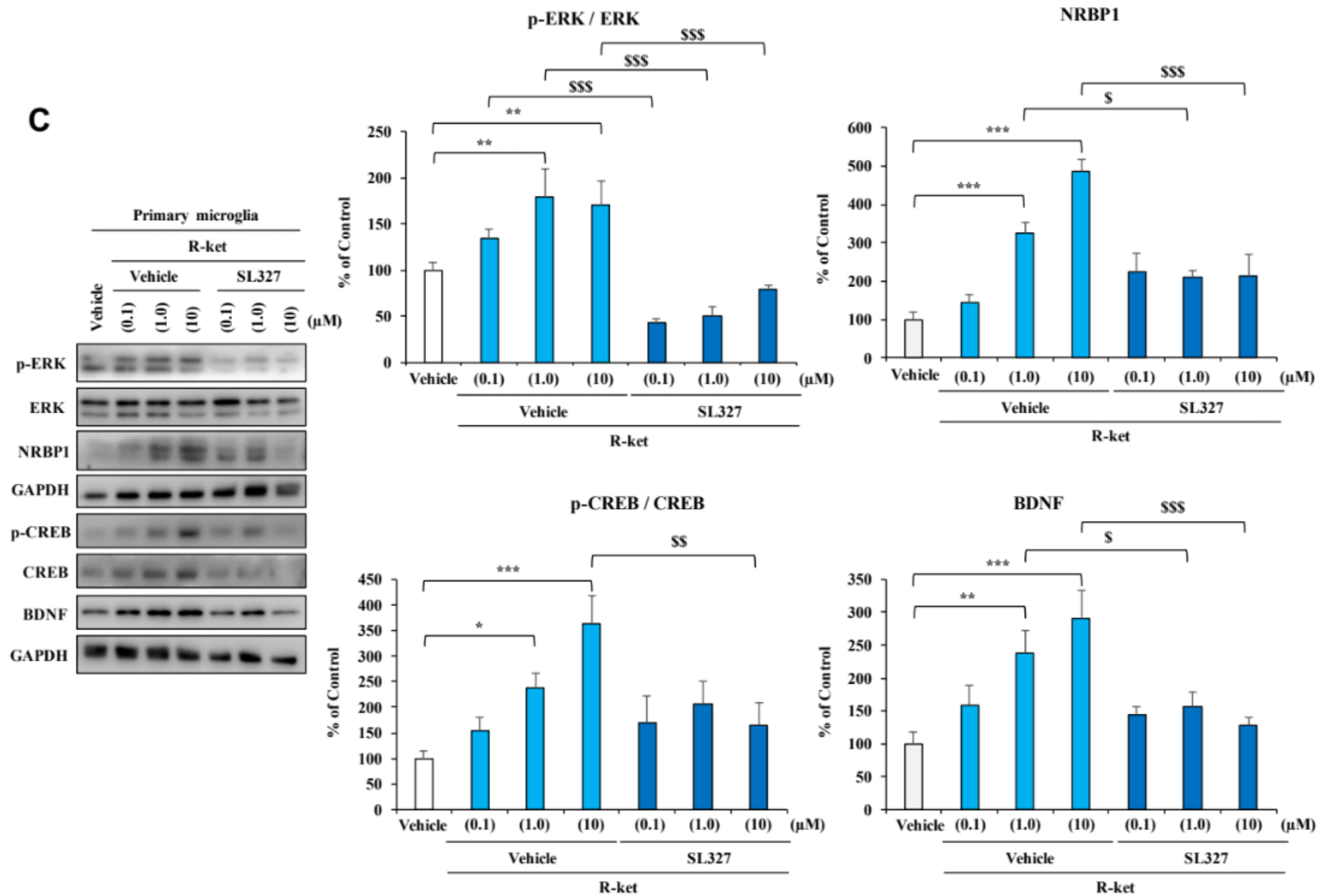
**Figure S3A:** Expression of NRBP1 and the ratio of p-CREB/CREB in BV2 cells treated with siRNA-NRBP1. The data are the mean  $\pm$  SEM (n = 4). \*\* $P$  < 0.01, \*\*\* $P$  < 0.001, \$ $P$  < 0.05, \$\$ $P$  < 0.01, \$\$\$ $P$  < 0.001 (one-way ANOVA).

The BV2 cell line was obtained by immortalization of mouse brain microglial cells with a retrovirus carrying the v-raf/v-myc oncogenes. BV2 cells mimic the morphological (small cell body with branched processes) and functional characteristics of primary microglia, retaining their ability to produce cytokines and respond to stimuli like lipopolysaccharide (LPS) (inducing inflammation).

**Figure S3B,C GROUPE 3**

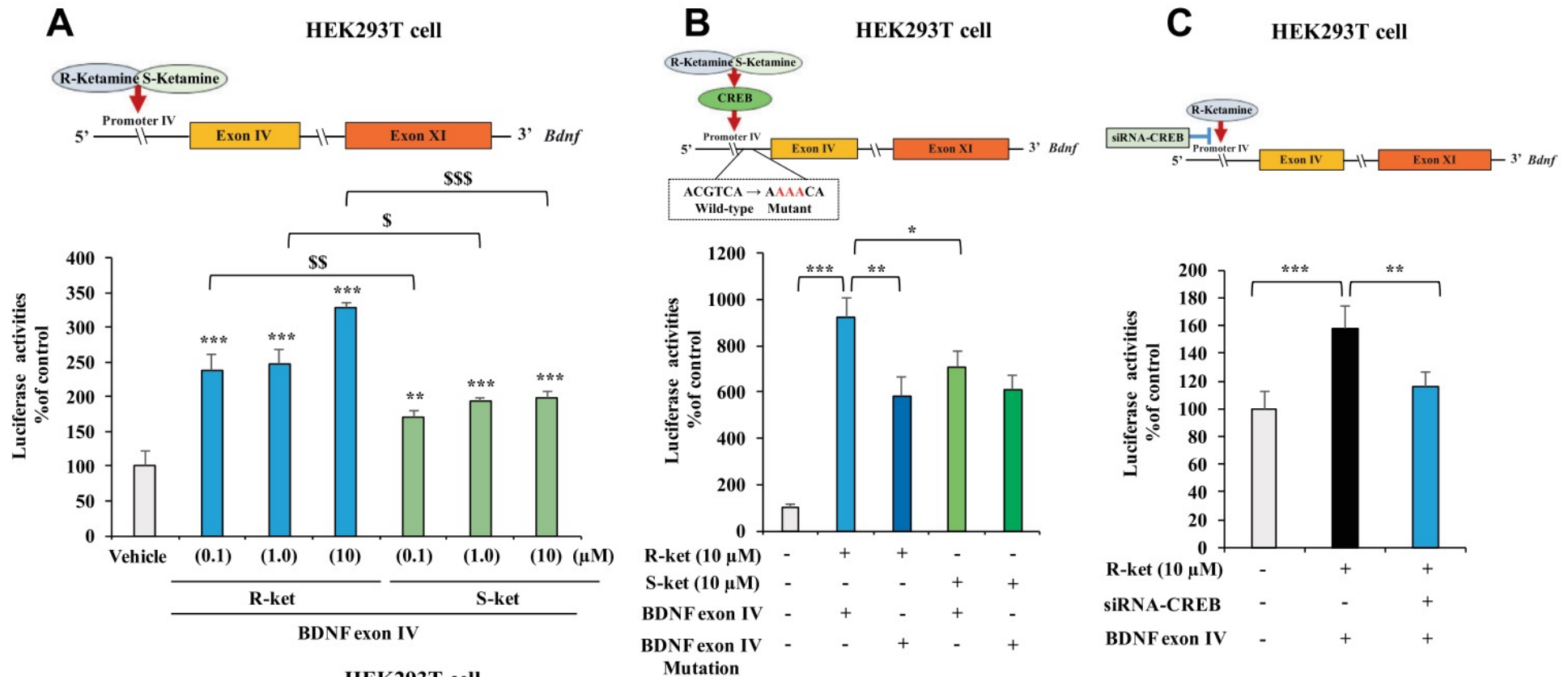


**Figure S3B:** Expression of NRBP1 and the ratio of p-CREB/CREB in the primary microglia by (*R*)-ketamine or (*S*)-ketamine. The data are the mean  $\pm$  SEM ( $n = 4$ ). \*\* $P < 0.01$ , \*\*\* $P < 0.001$ , \$ $P < 0.05$ , \$\$\$ $P < 0.001$  (one-way ANOVA).



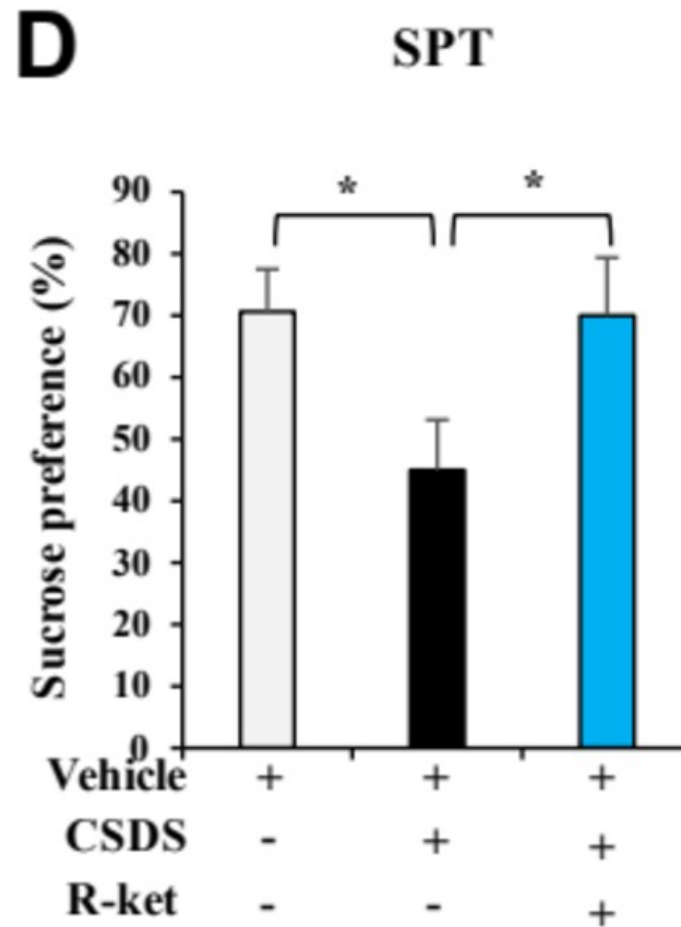
**Figure S3C:** Expression the ratio of p-ERK/ERK, NRBP1, the ratio of p-CREB/CREB and BDNF in the primary microglia by (*R*)-ketamine with or without SL327 (ERK inhibitor). The data are the mean  $\pm$  SEM ( $n = 6$ ).  $**P < 0.01$ ,  $***P < 0.001$ ,  $*P < 0.05$ ,  $$$P < 0.01$ ,  $$$$P < 0.001$  (one-way ANOVA).

**Figure 1 A-C GROUPE 4**



**Figure 1A-C:** **A.** Effects of ( R )-ketamine and ( S )-ketamine on the BDNF activation. **A, B** The luciferase assay for BDNF exon IV promoter. **A** BDNF exonIV promoter activity in the HEK293T cells treated with (R)-ketamine (0.1, 1.0, and 10 μ M) or (S)-ketamine (0.1, 1.0, and 10 μ M). The data are the mean ± SEM (n = 6). \*\*P < 0.01; \*\*\*P < 0.001 compared to vehicle group. \$ P < 0.05; \$\$P < 0.01; \$\$\$P < 0.001 (one-way ANOVA). **B.** BDNF exon IV promoter activity in the HEK293T cells treated with (R)-ketamine (10 μ M) [or (S)-ketamine (10 μ M)] with or without mutation (Mut) plasmids. The data are the mean ± SEM (n = 9 or 10). \*\*P < 0.01; \*\*\*P < 0.001 (one-way ANOVA). **C.** BDNF exon IV promoter activity in the HEK293T cells treated with (R)-ketamine (10 μ M) with or without siRNA-CREB plasmids. The data are the mean ± SEM (n = 4). \*\*P < 0.01; \*\*\*P < 0.001 (one-way ANOVA).

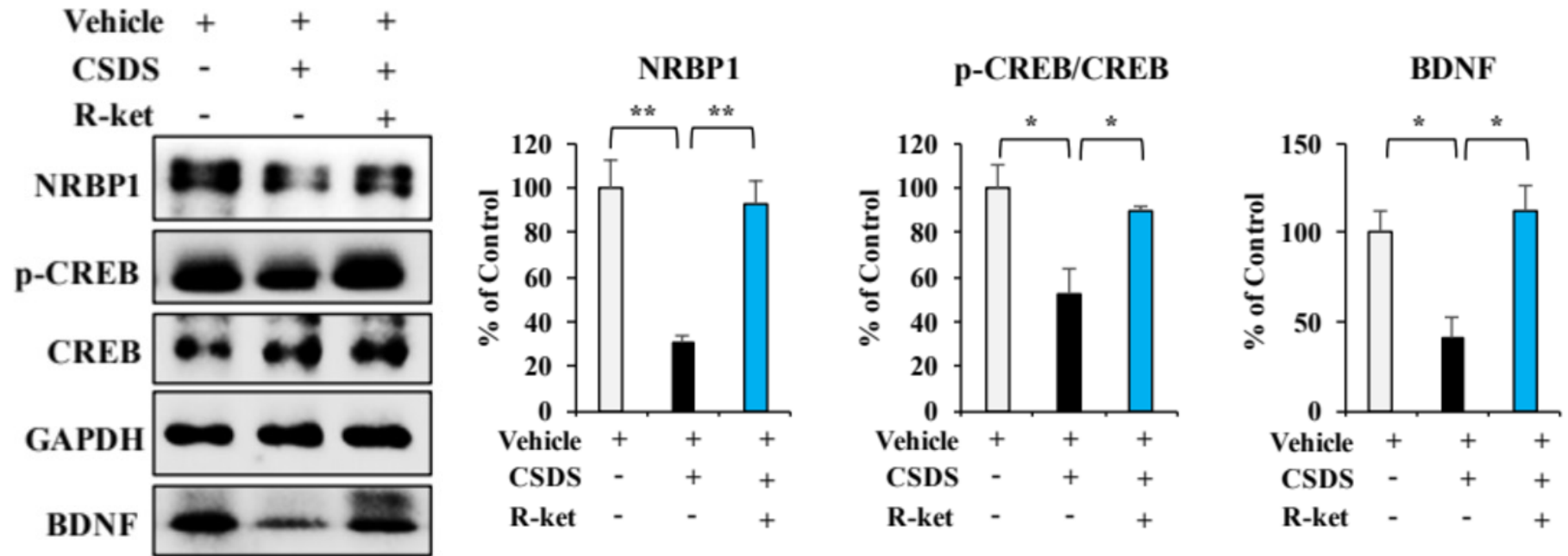
Figure S7D-F GROUPE 5



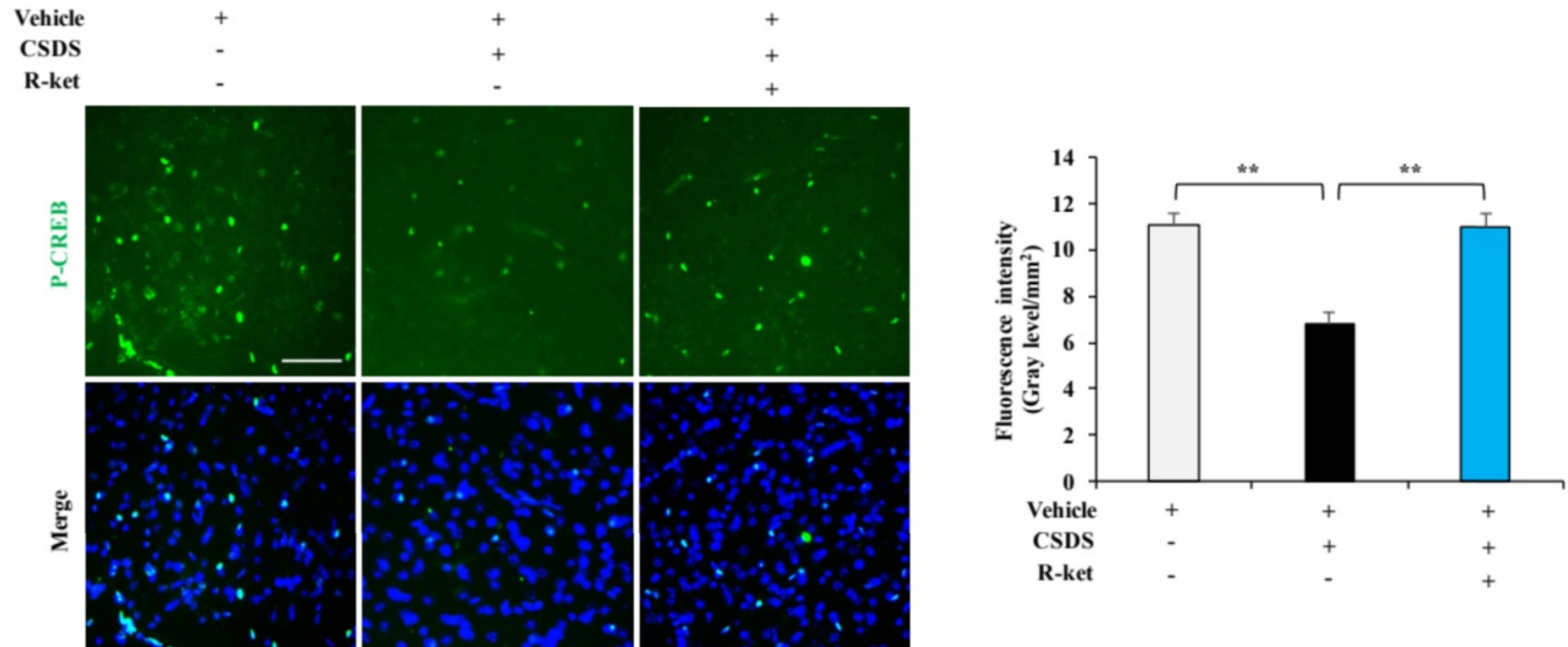
**Figure S7D:** Sucrose Preference Test (SPT).. The data are the mean  $\pm$  SEM ( $n = 8$  or  $9$ ).  $*P < 0.05$  (one-way ANOVA).

The sucrose preference test (SPT) is a behavioral test used in rodents to assess anhedonia, a symptom of depression characterized by a reduced ability to experience pleasure. The test involves a two-bottle choice paradigm where a rodent is presented with plain water and a sucrose solution in its home cage. A lower preference for the sucrose solution compared to water over a set period is interpreted as anhedonia, indicating a diminished hedonic response .

**E**



**Figure S7E:** Expression of NRBP1, the ratio of p-CREB/CREB and BDNF in the Prefrontal cortex. The data are the mean  $\pm$  SEM (n = 5). \* $P$  < 0.05, \*\* $P$  < 0.01 (one-way ANOVA)

**F**

**Figure S7F:** The immunofluorescence analysis for p-CREB in the mPFC. The data are the mean  $\pm$  SEM ( $n = 5$ ).  $**P < 0.01$  (one-way ANOVA). Scale bar = 50  $\mu\text{m}$ .